Lab 3: Component Modelling & Architectural Pattern

Selection Objective

Evaluate different architectural styles, select the most appropriate one for the assigned scenario, and create a comprehensive UML Component Diagram showing modules, interfaces, and dependencies. This lab focuses on understanding how components interact within a system and how architectural decisions impact system design.

Duration

60 minutes

Software Requirements

- UML tool (draw.io, LucidChart, StarUML or any online UML tool)

Learning Outcomes for Lab 3

By the end of this lab, you will be able to:

1. **Evaluate architectural styles:**
    - **Compare and contrast different architectural patterns (Layered, Microservices, Client-Server)**
    - **Analyse pros and cons of each pattern relative to specific scenarios**
2. **Select appropriate architecture:**
    - **Choose the most suitable architectural style based on system requirements ● Justify architectural decisions with concrete reasoning**
3. **Design component diagrams:**
    - **Create UML component diagrams with proper notation**
    - **Show provided and required interfaces between components**
    - **Demonstrate understanding of component dependencies**
4. **Document architectural decisions:**
    - **Write clear, technical justifications for architectural choices**
    - **Explain security and performance implications**

Introduction:

Component modelling Fundamentals

What is a Component?

A component is a replaceable and executable piece of a system whose implementation details are hidden. Key characteristics:

- **Modular: Encapsulates specific functionality**
- **Replaceable: Can be swapped with other components that provide the same interface ●**

Black Box: External behavior is defined by interfaces, internal implementation is hidden

- **Executable: Participates in system execution**

Component Diagram Elements

1. Component Notation



- **Represented as a rectangle with <<component>> stereotype**
- **Component name placed at center**
- **Optional component icon in upper-right corner**
- **Implementation details are hidden from external view**

2. Interfaces

Provided Interface (Ball notation):



- **Services that a component offers to other components**
- **Shown as a circle connected to the component**
- **Represents what the component can do for others**

Required Interface (Socket notation):

- **Services that a component needs from other components**
- **Shown as a semicircle connected to the component**
- **Represents what the component needs from others**

3. Ports



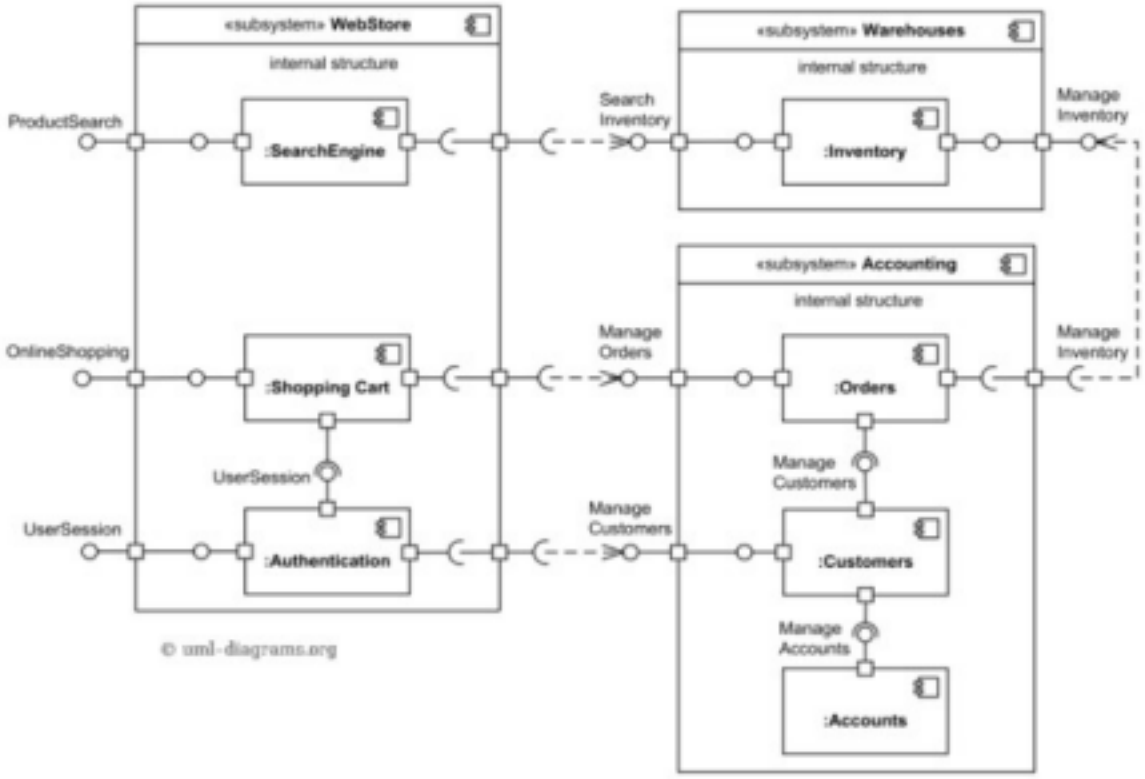
- **Interaction points between components and external environment**
- **Group related provided and required interfaces**
- **Can be public (drawn on boundary) or private (drawn inside)**

4. Dependencies

- **Usage Dependencies: Shown as dashed arrows with <<use>> stereotype**
- **Assembly Connectors: Connect provided interfaces to required interfaces**



Example Component diagram:



Online shopping UML component diagram example with three related subsystems - WebStore, Warehouses, and Accounting.

Architectural Styles Overview

1. Layered Architecture

- **Structure: Organized in horizontal layers (Presentation, Business, Data) • Pros: Clear separation of concerns, easy to understand, good for traditional applications**
- **Cons: Performance overhead, tight coupling between layers, difficult to scale individual components**

2. Microservices Architecture

- **Structure: Small, independently deployable services**
- **Pros: Independent scaling, fault isolation, technology diversity, independent deployment**
- **Cons: Operational complexity, network latency, data consistency challenges 3.**

Client-Server Architecture

- **Structure: Centralized server with multiple clients**
- **Pros: Centralized control, simple deployment, easy data consistency**

- **Cons: Single point of failure, scalability bottleneck, limited fault tolerance**

Example Scenario:

Self-Service Coffee Kiosk System

You are designing the software architecture for a Self-Service Coffee Kiosk system in a busy café. The kiosk must provide a complete coffee ordering experience with the following requirements:

Core Functionality:

- Customers can select from 3 coffee types (Espresso, Americano, Latte) •

  Customers can choose from 2 drink sizes (Small, Large)

- Customers can pay via credit card only

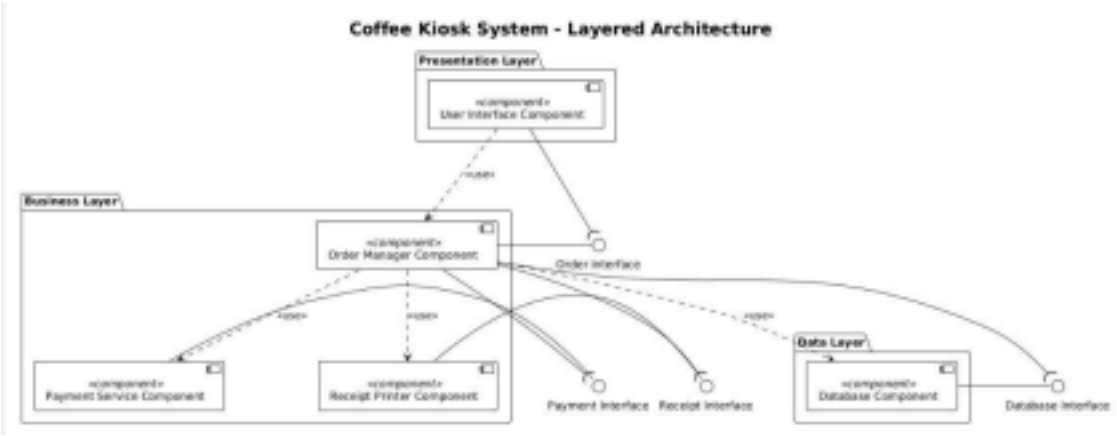
- System prints receipts with order details

Technical Constraints:

- Must handle touch screen interface interactions

- Must connect to receipt printer hardware

- Must store menu data and pricing information



Deliverables

1. Component Diagram (PNG or PDF)

Your diagram must include atleast 5 components (Adding attributes is optional).

Interface Requirements:

- Show atleast 4 interfaces between components:
- Use proper UML notation for provided/required interfaces
- Show data flow and component interactions

2. Written Justification (Word document, max 1 page)

Lab 3 Steps

Step 1: Scenario Review

- Read the assigned scenario thoroughly
- Identify key functional and non-functional requirements
- List the main challenges (performance, usability, security)

Step 2: Architectural Style Analysis

Analysis: Analyse each architectural style.

Step 3: Architecture Selection & Component Identification

Select Architecture

Identify Components: Based on your chosen architecture, identify all necessary components *(Two components are given below, please identify the remaining 3):*

1. Order Manager Component
2. Payment Service Component

Step 4: Component Diagram Creation

Setup:

- Open your UML tool (draw.io recommended)
- Set up the canvas with adequate space

Draw Components:

- Create component rectangles for each of the 5 identified components •

  Use <<component>> stereotype or clear component notation

- Arrange components logically on the canvas

- Group related components (e.g., customer-facing, data storage)

Add Interfaces: Define and draw the required interfaces between components:

Key Interfaces to Show:

Given:

- Order Manager ↔ Payment Service: Payment processing requests

Interface Notation Guidelines:

- Use solid lines for internal service communications

- Label interfaces with protocol/technology (API calls, hardware interfaces, database queries)

- Use provided interface (ball) and required interface (socket) notation where appropriate

Create a one-page Word document with the following structure:

Architecture Selection: "We chose [Architecture] for the <Assigned> System."

Your justification must include:

- Architectural Choice: State which architectural style you selected • Two Reasons:

  Provide two specific, scenario-related reasons for your choice • Security Advantage:

  Explain one way your architecture addresses security concerns

- Performance Benefit: Describe one performance advantage of your chosen architecture

Step 6: Diagram Finalization & Export

- Review diagram for completeness

- Ensure all required components are present
- Verify interface directions and labels are correct
- Check UML notation compliance
- Export as PNG or PDF
- Save justification document as PDF
- Submit the same through the Github repo (Ex: Lab3 under SELABS)